

Solution Requirements

Date	01 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	4 Marks

Functional Requirements

FR No.	Functional Requirement	Description
FR-1	User Registration / Access	Analysts and applicants can access the web application to submit applicant details
FR-2	Data Input Form	System accepts income, employment, credit history, and demographic details through a form
FR-3	Data Preprocessing	System cleans, encodes, and scales input data before feeding it to the model
FR-4	Model Prediction	System predicts approval/rejection using the best-performing trained classification model
FR-5	Risk Categorization	System assigns a risk category (low/medium/high) alongside the approval decision
FR-6	Result Display	System displays the prediction instantly on the web interface
FR-7	Batch Screening	Compliance officers can upload a batch of applicants for bulk screening
FR-8	Cloud Deployment	The trained model is deployed via IBM Watson Machine Learning for scalable access

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The web interface should be simple and intuitive for both technical and non-technical users
NFR-2	Performance	The system should return a prediction within a few seconds of submission
NFR-3	Reliability	The model should give consistent predictions for the same input data
NFR-4	Security	Applicant financial data should be handled and stored securely
NFR-5	Scalability	The system should support increasing numbers of concurrent prediction requests via cloud hosting
NFR-6	Availability	The web application should be accessible with minimal downtime
NFR-7	Maintainability	Code should be modular so the model can be retrained or swapped without major rework

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